

ABHIJIT BENDALE

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EXECUTIVE SUMMARY

Applied AI leader and hands-on engineer with 15+ years building and shipping large-scale ML systems across LLMs, multimodal AI, and computer vision. Delivered production AI platforms at global scale (400M+ users and 100M+ devices). Founder & CTO of Two Platforms (raised 20M seed, 10M ARR), where I built multilingual LLM and multimodal AI platforms deployed at national scale. Former CTO of Samsung STAR Labs, where I built NEON, a real-time generative digital human platform later integrated into Samsung Electronics. Previously at Samsung Electronics, architected on-device AI platforms deployed across 100M+ Galaxy devices.

Deep expertise in large-scale ML systems, distributed training, post-training, RLHF, evals, Agentic AI systems. Expertise in performance optimization across model, runtime, and hardware, with a focus on low-latency and high-throughput AI systems. Inventor on 30+ patents with peer-reviewed papers at leading AI conferences (3800+ citations).

EXPERTISE & TECHNICAL SKILLS

- **Foundation Models & Training:** LLM/VLM pre-training and fine-tuning; Transformers, MoE, CLIP-style multimodal models; distributed training with DeepSpeed, Ray, FSDP.
- **Post-Training & Alignment:** SFT, RLHF, preference optimization, reward modeling, TRL, PEFT, LLM-as-judge evaluation, human-in-the-loop pipelines.
- **Inference & Serving:** High-throughput LLM serving with vLLM, SGLang, NVIDIA NIM, TorchServe; KV-cache optimization, quantization.
- **Retrieval & Search Systems:** Billion-scale vector search, FAISS, Vertex Matching Engine, PgVector, RAG pipelines; multimodal retrieval across text, image, audio, and video.
- **AI Agents & Orchestration:** Tool-using agents, function calling, MCP, agent memory, multi-agent orchestration, agent evals & observability; LangGraph, PydanticAI, Google ADK, Agent SDKs.
- **Cloud & Infrastructure:** Production AI platforms on AWS, GCP. Compute and orchestration (EKS, ECS, GKE), ML platforms (SageMaker, Vertex AI), data and storage (RDS, DynamoDB, AlloyDB, Redis), Cloudflare, streaming and observability.
- **Languages & Core Tools:** C++, Python, TypeScript, PyTorch, CUDA/Triton, NCCL.

WORK EXPERIENCE

Two Platforms

2021 – 2025

Founder & Chief Technology Officer

Campbell, CA

Founded Two Platforms - AI infrastructure & applications startup (\$20M seed from Jio Platforms & Naver Corp.) focused on multilingual LLMs, multimodal/voice AI, and enterprise applications for the APAC market. Scaled team of 20+ engineers across ML, platform and product.

- Built SUTRA, a multilingual AI platform spanning LLMs, VLMs, semantic search, and voice AI. Developed the full model lifecycle from data collection, pre-training, SFT, RLHF, evaluation and production deployment.
- Built and operated 200+ GPU AI infrastructure for distributed model training and high-throughput inference, including model serving, observability and performance optimization for production LLM and multimodal workloads.
- Deployed enterprise AI search, multi-agent workflows, and voice AI agents with strategic customers including Jio Platforms across cloud, VPC, on-prem, and regulated environments, generating \$10M+ ARR and powering multimodal AI experiences across text, image, audio, video, and documents for 400M+ users.
- Built consumer AI products including ZAPPY (AI chat, 1M+ users), AutoCameo (personalized video messaging), and SUTRA Avatar (AI video calls), launched nationally in partnership with Jio Platforms.

Samsung Technology & Advanced Research (STAR) Labs
Chief Technology Officer

2019 - 2021
Mountain View, CA

As CTO of STAR Labs, led NEON, a real-time generative digital-human platform, from research prototype to commercialization; later integrated into Samsung Electronics at a \$50M+ internal valuation. Built and led a 40+ person multidisciplinary team across AI research, platform, product.

- Architected NEON, an interactive avatar platform for creating 4K photorealistic digital humans with sub-100 ms latency. Developed the core audio-to-video neural rendering stack and launched it at CES 2020.
- Designed and trained the core real-time multimodal generative AI stack, combining generalized speech-to-motion models, identity personalization, neural rendering, optimized GPU inference, and streaming media delivery.
- Built a production ML and media platform spanning data pipelines, model training and evaluation, low-latency serving, observability, SDKs, cloud deployment, and enterprise integration.
- Drove commercialization through NEON View, NEON Frame, and NEON Studio for interactive customer service, digital displays, mobile and web avatars, and enterprise content generation.
- Drove commercialization via adoption by Samsung Display/Mobile and APAC enterprise partners in automotive, telecom, media, and finance (e.g., Hyundai, SK Telecom, Shinhan Bank, CJ ENM etc.).

Samsung Research America
Director of AI / Principal Engineer / Staff Engineer

2016 - 2019
Mountain View, CA

Led architecture and scaling of on-device AI and perception platforms deployed across 100M+ Samsung Galaxy devices. Built and managed a team of 60+ engineers and researchers (20+ PhDs) with a \$15M+/year annual R&D budget.

Bixby Vision & On-Device AI

- Built and scaled Bixby Vision, on-device AI deployed across 100M+ Galaxy devices, enabling visual search, translation, shopping, and scene understanding.
- Built Samsung's on-device AI platform from first principles, enabling ML deployment across Android, Tizen, Galaxy devices, Smart TVs, digital displays, and XR systems under strict latency, power, and memory constraints.
- Designed real-time camera and perception pipelines, processing live video frames for object detection, OCR, scene understanding, and AR experiences with low-latency on-device inference.
- Built platform SDK and APIs (Vision, AR, rendering) enabling partner applications and services to integrate with Bixby Vision, creating an extensible ecosystem.
- Reduced latency and power consumption across Galaxy devices by leading hardware-software co-design with Samsung Semiconductor (Exynos NPU), Qualcomm, and Google.
- Drove transition from applied research to production systems, defining architecture, evaluation, and deployment strategies for large-scale commercialization.

AR & Mixed Reality Platforms

- Led AR and Mixed Reality platform initiatives in collaboration with Microsoft (HoloLens) and Google (Android / ARCore), shaping hardware-software integration and platform architecture.
- Directed and contributed hands-on to Samsung's in-house AR/MR stack, building SLAM and sensor fusion systems using RGB, event-based vision (DVS), IMU, and depth sensors for real-time spatial mapping.
- Developed low-latency, power-efficient perception and rendering pipelines for XR, including patented approaches for event-driven vision and sensor fusion.

RESEARCH EXPERIENCE

SRI International

PhD Intern

June 2014 – September 2014

Princeton, NJ

- Built algorithms to automatically summarize and reconstruct sequential storylines from large-scale photo collections, modeling temporal structure across millions of images DARPA ALADDIN.

IDIAP Research Institute (EPFL)

Visiting Student (with Prof. Sebastien Marcel)

October 2013 – January 2014

Switzerland

- Researched continual learning and online model adaptation for large-scale biometric recognition under distribution shift (EU / NSF Open Vision).

MIT Media Lab

Research Scientist (with Prof. Ramesh Raskar)

August 2010 – July 2011

Cambridge, MA

- Developed VisionBlocks, a social computer vision platform (Google App Engine Award), and conducted image super-resolution research for DARPA ARGUS-IS.

MIT McGovern Institute & Dept. of Brain & Cognitive Sciences

Visiting Student & Research Scientist (with Prof. James DiCarlo)

July 2009 – July 2010

Cambridge, MA

- Developed deep learning and biologically inspired vision models for object recognition and pedestrian detection. Research supported by DARPA NeoVision.

PUBLICATIONS & PATENTS

Selected list of publications (cited 3800+ times) is listed below:

1. A. Bendale *et.al.* "SUTRA: Scalable Multilingual Language Model Architecture" 2024 (*arxiv* 2405.06694)
2. A. Bendale, T. Boulton "Towards Open Set Deep Network" CVPR 2016 (oral, 2.5K+ citations)
3. A. Bendale, T. Boulton "Towards Open World Recognition" CVPR 2015 (oral, 900+ citations)
4. A. Bendale, T. Boulton "Reliable Posterior Probab. Estimation for Face Recognition" Biometrics W, CVPR 2014
5. A. Bendale, *et.al.*, R. Raskar "VisionBlocks: Social Computer Vision Platform" SocialCom 2011

Selected list of Patents (from 30+):

1. Generating digital avatar (US 11544886, 2023)
2. Real-time interactive digital agents (US 17686201, 2023)
3. CMOS-assisted inside-out DVS tracking (US 11381741, 2023)
4. Semi-dense depth estimation with DVS (US 11143879, 2021)
5. Semantic mapping for low-power AR (US 10812711, 2020),

Full list is available via Google Scholar.

EDUCATION

University of Colorado

Ph.D. in Computer Science (Dissertation: Open World Recognition)

Advisor: Prof. Terrance Boulton

2011 – 2015

University of Colorado

M.S. in Computer Science

2007 – 2009

University of Pune

B.E. in Electronics and Telecommunications

2003 – 2007